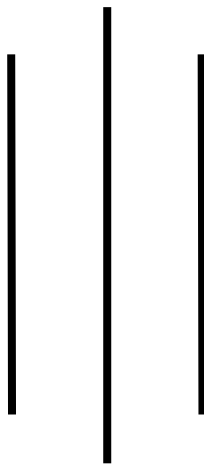


TRIBHUVAN UNIVERSITY

PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR



MANAGEMENT INFORMATION SYSTEM (BIT 353)

SUBMITTED BY:

NAME: SURESH DAHAL

ROLL: 23

CLASS: BIT-III/II

DATE: 2083/04/22

SUBMITTED TO:

SHIB KUMAR MAHATO

CHECKED BY

1. Familiarization of Data Visualization by using Microsoft Excel.

THEORY

Data visualization is the practice of translating raw information into graphical formats, such as charts, graphs, and maps, to make data insights more accessible. In the context of Management Information Systems, data visualization is a critical component of Business Intelligence. It allows management to monitor Key Performance Indicators (KPIs), identify operational trends, and make evidence-based decisions without having to parse through thousands of rows of raw data.

Microsoft Excel serves as a foundational tool for data visualization due to its capability to handle data cleaning, pivot reporting, and dynamic charting. By utilizing features like conditional formatting, pivot tables, and various chart types (bar, line, pie), users can transform unstructured data into meaningful visual assets that communicate business performance effectively.

PROCEDURE

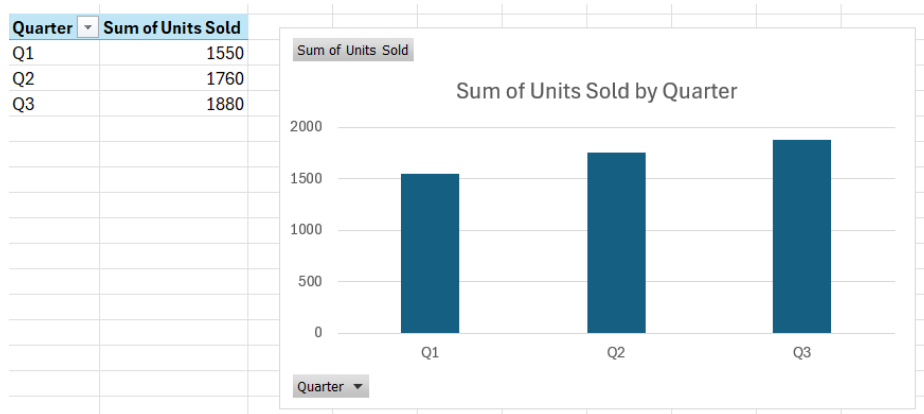
- A dataset containing relevant organizational information was prepared and organized within an Excel spreadsheet.

Region	Product	Units Sold	Revenue	Quarter
North	Hardware	450	45000	Q1
South	Software	300	60000	Q1
East	Hardware	520	52000	Q1
West	Software	280	56000	Q1
North	Hardware	480	48000	Q2
South	Software	320	64000	Q2
East	Hardware	610	61000	Q2

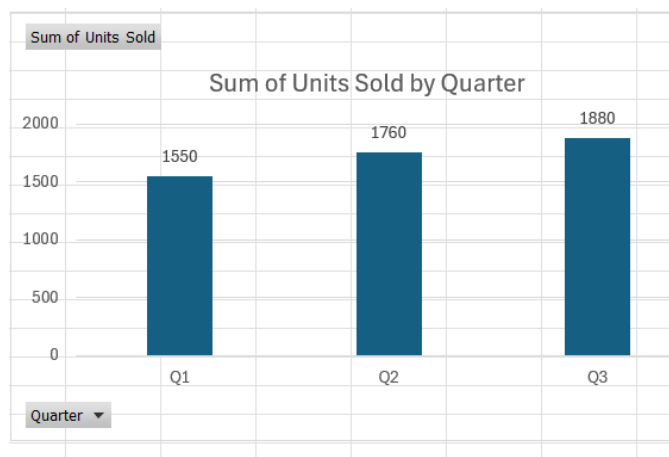
- Conditional formatting rules were applied to numerical columns to visually distinguish high and low-performance values.

Region	Product	Units Sold	Revenue	Quarter
North	Hardware	450	45000	Q1
South	Software	300	60000	Q1
East	Hardware	520	52000	Q1
West	Software	280	56000	Q1
North	Hardware	480	48000	Q2
South	Software	320	64000	Q2
East	Hardware	610	61000	Q2

- A range of data was selected, and the “Insert” tab was utilized to generate Clustered Column Charts for comparative analysis.



- Chart data label was configured to ensure the clarity and readability of the information.

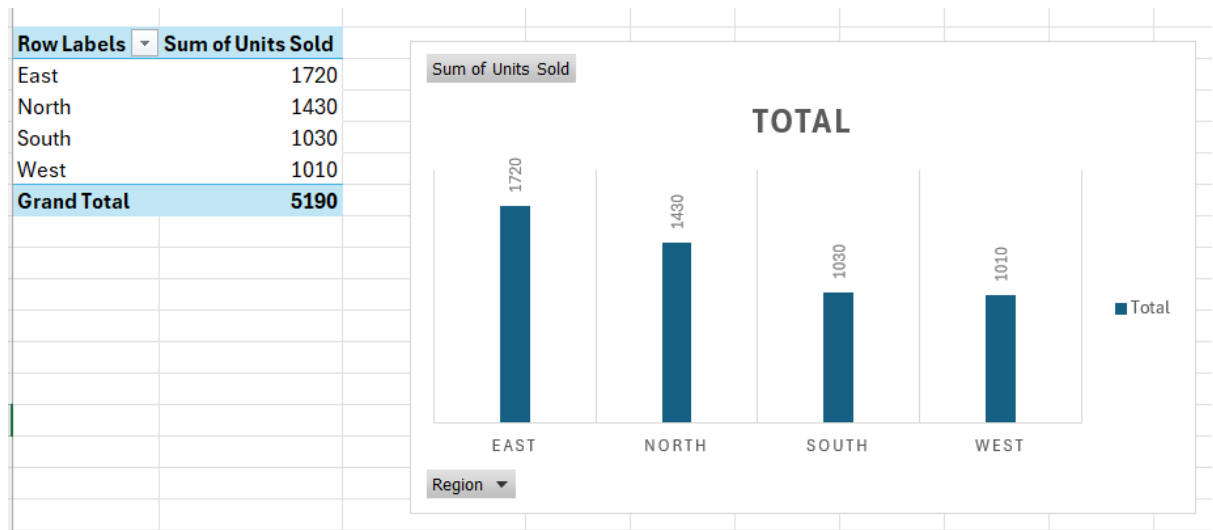


- A PivotTable was created by selecting the dataset and navigating to “Insert > PivotTable.” The '**Region**' field was added to the **Rows** area, and the '**Units Sold**' field was added to the **Values** area to visualize the number of units sold per region.

Row Labels	Sum of Units Sold
East	1720
North	1430
South	1030
West	1010
Grand Total	5190

- A PivotChart was linked to the PivotTable to facilitate an interactive visual summary, allowing for real-time adjustments as the source data was filtered
 - With the PivotTable selected, the PivotTable Analyze tab was accessed from the top ribbon.

- The PivotChart function was selected, and a Clustered Column Chart was chosen to visualize the data.
- The chart was then configured to provide an interactive summary, where the filter buttons on the chart were used to dynamically update the data display, demonstrating the synchronization between the PivotTable and the PivotChart.



CONCLUSION

This lab successfully demonstrated the application of Excel's visualization tools, such as conditional formatting, PivotTables, and charts to transform raw data into actionable insights. These techniques proved essential for summarizing business metrics and facilitating data-driven decision-making, effectively meeting the lab's objective.

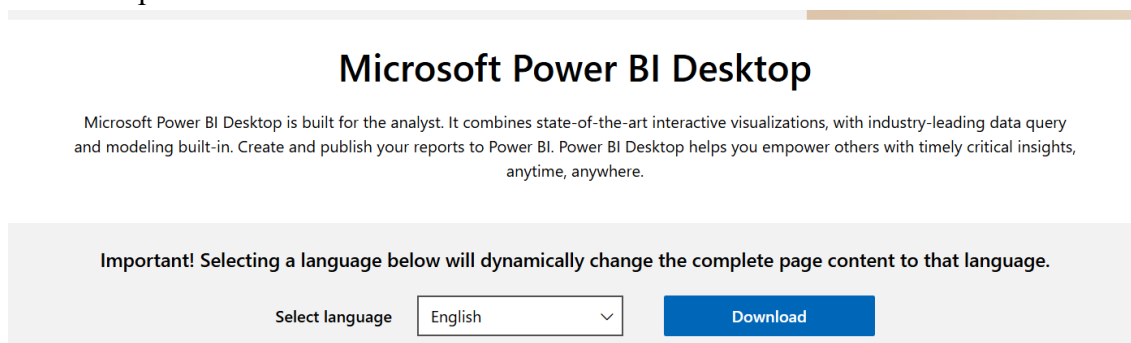
2. Installation Procedure of Microsoft Power BI Desktop

THEORY

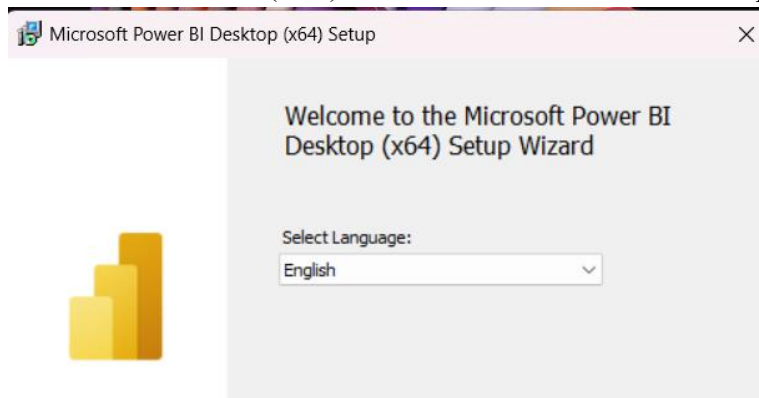
Microsoft Power BI Desktop is a specialized business intelligence application used for data connectivity, transformation, and visualization. As a core tool in the modern MIS toolkit, it allows users to connect to various data sources, perform complex data modeling, and develop interactive reports.

PROCEDURE

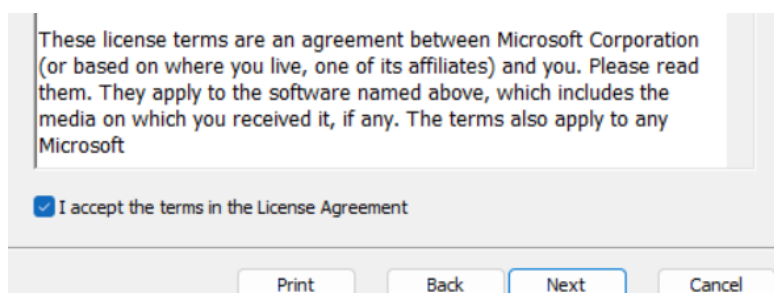
- The official Microsoft Power BI website was accessed, and the installer package for Power BI Desktop was downloaded.



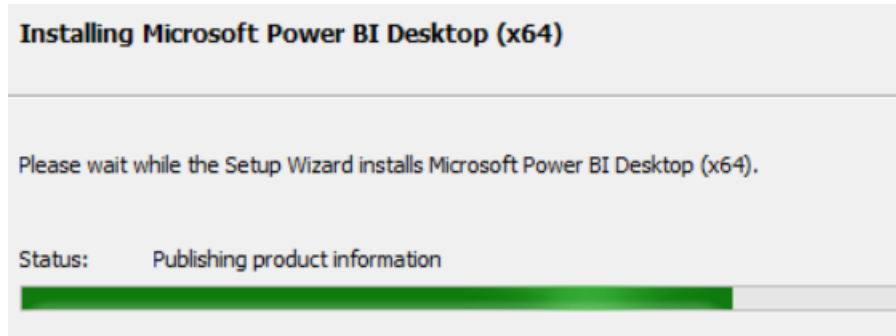
- The installation file (.exe) was executed to initiate the setup wizard.



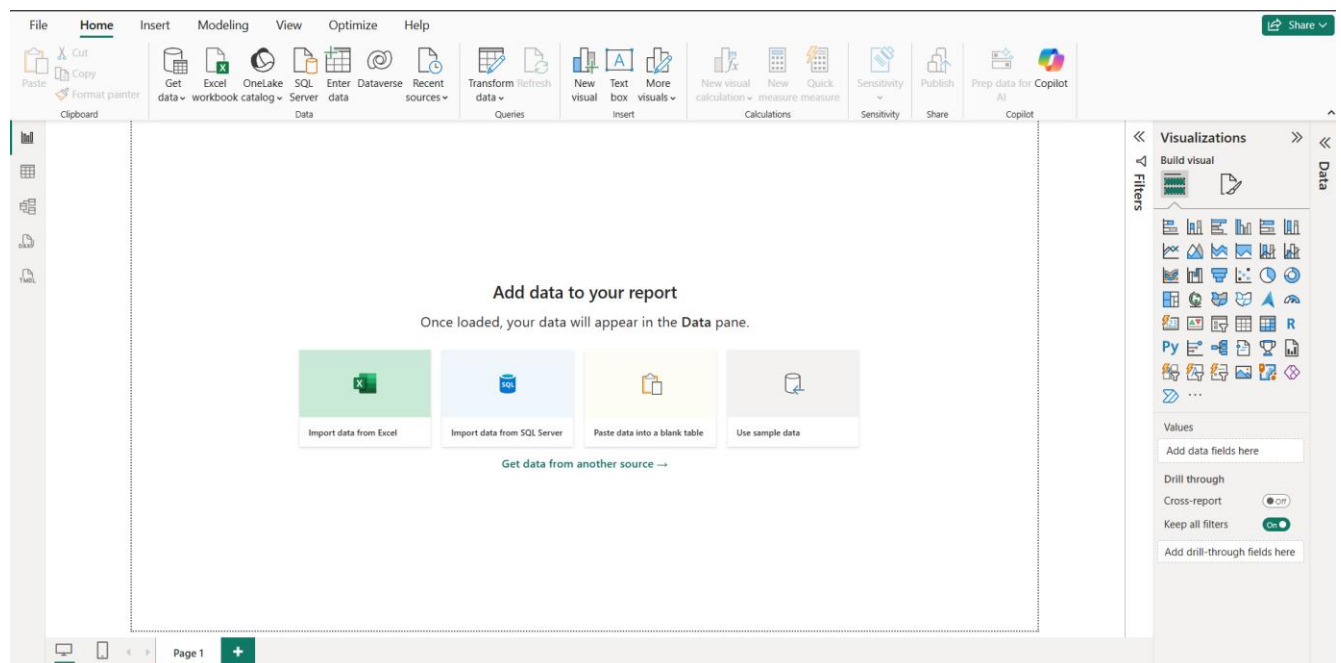
- The language preference was selected, and the setup process was followed by accepting the End-User License Agreement (EULA).



- The destination folder for the software installation was specified, and the installation process was completed by granting the necessary administrative permissions.



- Upon completion, the application was launched to verify the integrity of the installation and ensure the primary user interface was correctly initialized.



CONCLUSION

The installation of Microsoft Power BI Desktop was performed successfully. The application environment is now fully configured and ready for data import, transformation, and the creation of business intelligence reports.

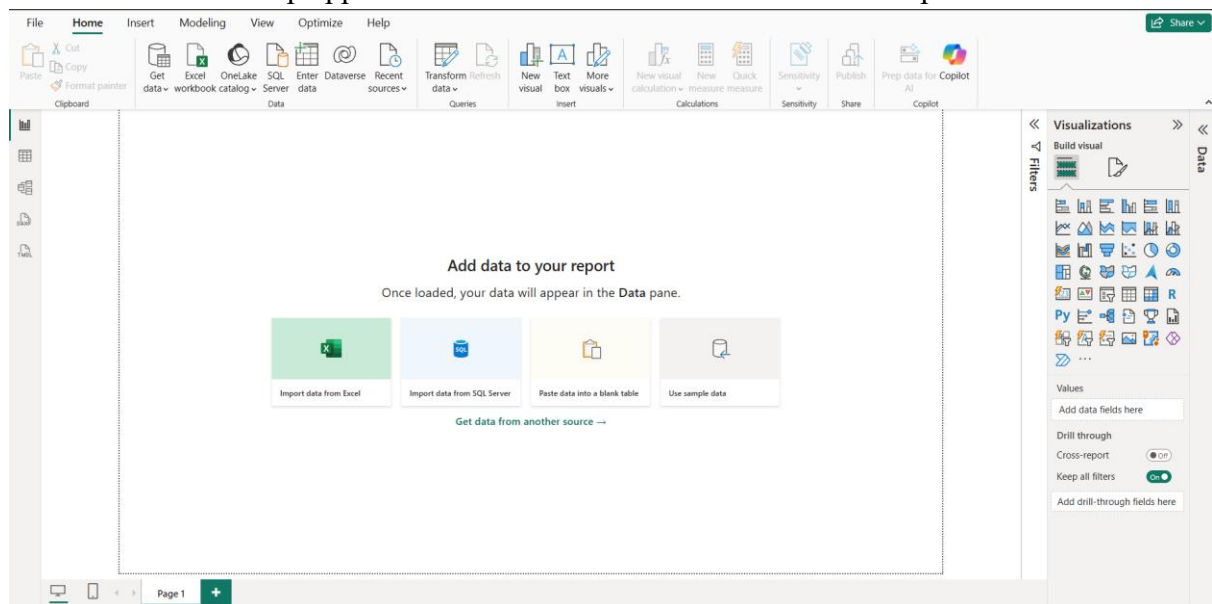
3. Introduction and Key Features of Power BI

THEORY

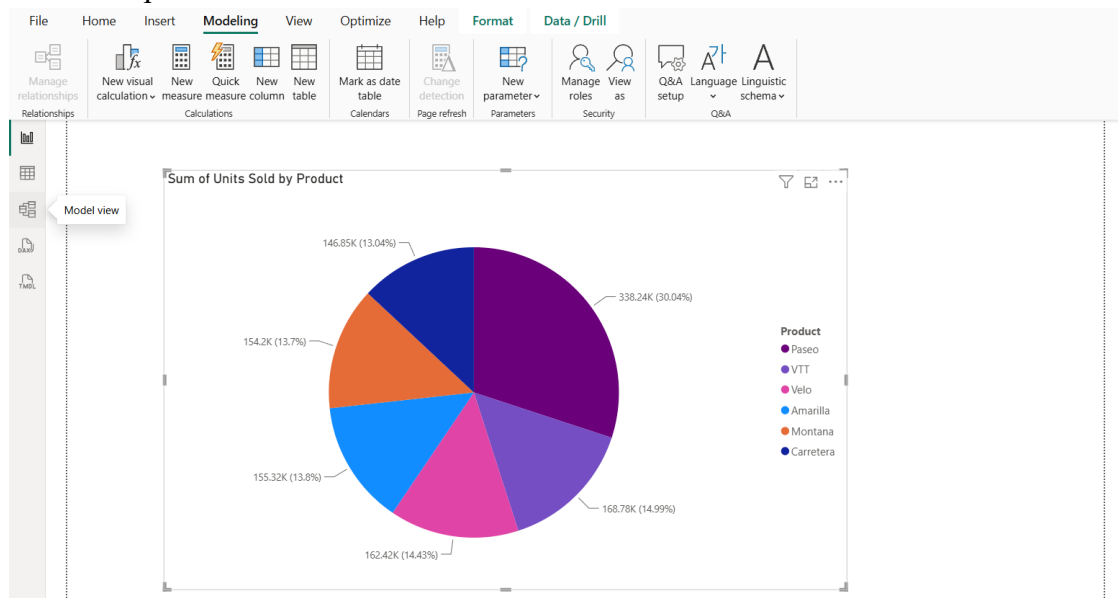
Microsoft Power BI is a comprehensive Business Intelligence (BI) platform that enables organizations to aggregate, analyze, and visualize data from diverse sources. In an MIS environment, Power BI acts as the bridge between raw data silos and strategic management insights. Its architecture is designed to simplify complex data modeling, enabling both technical and non-technical users to generate professional-grade reports and dashboards.

PROCEDURE

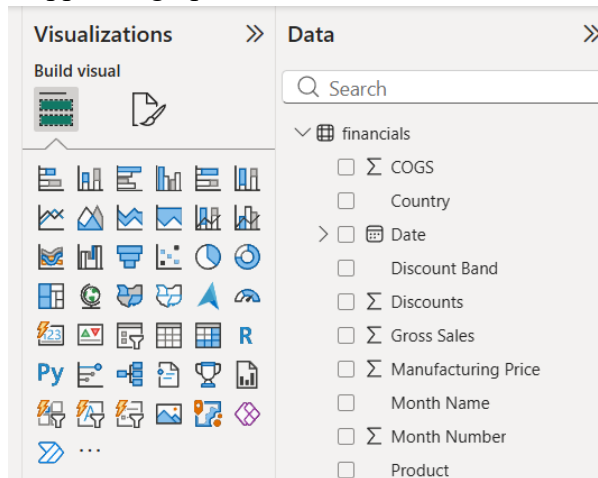
- The Power BI Desktop application was launched to initialize the workspace.



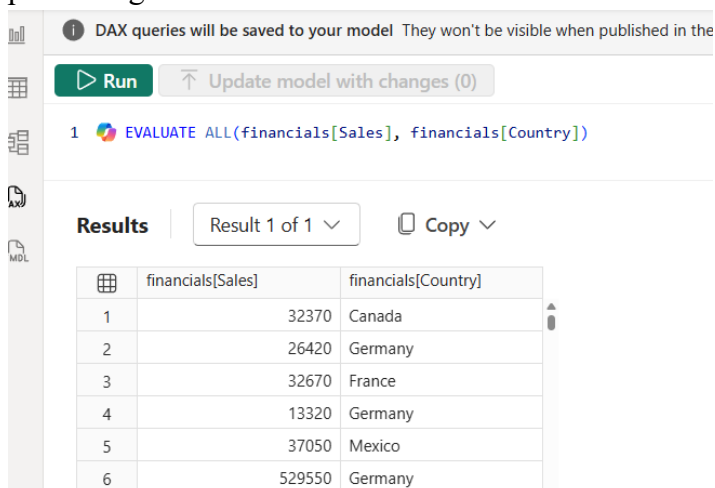
- The primary interface components were identified, specifically the **Report View** for visualization, the **Data View** for exploring raw tables, and the **Model View** for defining relationships between data entities.



- The **Visualizations Pane** and **Data Pane** were explored to understand how data fields are mapped to graphical elements.



- The application's capability to transform data via **Power Query** and perform calculations using **Data Analysis Expressions (DAX)** was examined to understand the backend processing flow.



CONCLUSION

This lab provided a practical introduction to the Power BI interface and its core analytical functions. By navigating its primary views and testing DAX expressions, it was confirmed that Power BI is a robust tool for aggregating and transforming raw data into meaningful business intelligence.

4. The Role of Power BI in Business Organizations

THEORY

Power BI serves as an essential Business Intelligence (BI) layer in Management Information Systems (MIS). By consolidating data from fragmented operational sources (e.g., finance, sales, and supply chain) into a unified dashboard, it allows organizations to monitor Key Performance Indicators (KPIs) in real-time. This capability shifts organizational decision-making from subjective intuition to objective, data-driven analysis, enabling managers to identify trends and address operational inefficiencies quickly.

PROCEDURE

- **Data Loading:** The "Financials" sample dataset was imported into the Power BI environment to simulate a real-world organizational database containing sales, cost, and profit information.

The screenshot displays the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, showing options like 'Manage relationships', 'New measure', 'Quick measure', 'New column', 'New table', and 'Mark as date table'. Below the ribbon, a data table named 'financials' is visible. The table has columns for Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit, and Date. The data includes various entries for different countries (Germany, Canada, France, Mexico, United States of America) and products (Carretera, Paseo, Velo, VTT, Amarilla).

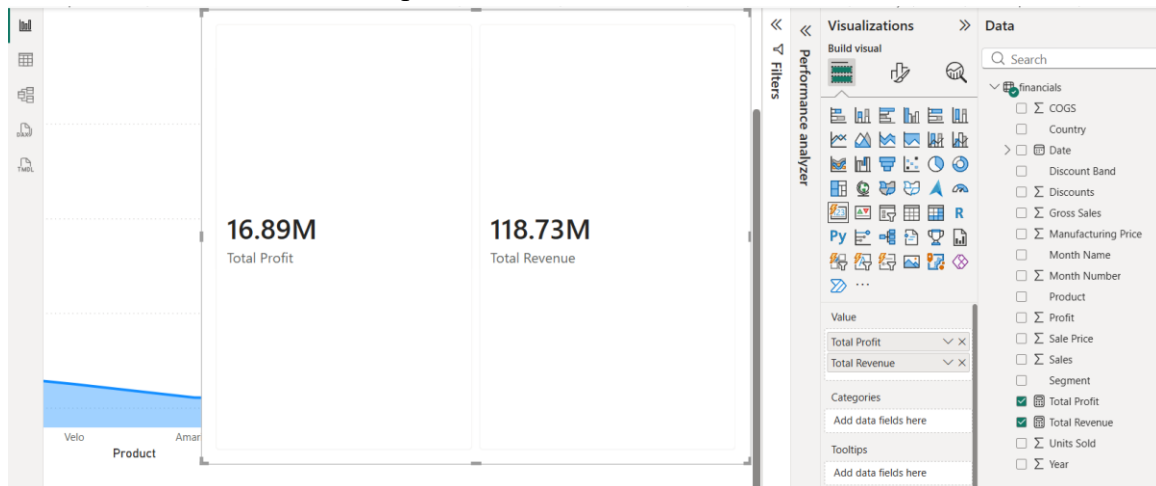
Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date
Government	Germany	Carretera	None	1513	3	350	529550	0	529550	393380	136170	Monday, December 1, 201
Government	Germany	Paseo	None	1006	10	350	352100	0	352100	261560	90540	Sunday, June 1, 201
Government	Canada	Paseo	None	1725	10	350	603750	0	603750	448500	155250	Friday, November 1, 201
Government	Germany	Paseo	None	1513	10	350	529550	0	529550	393380	136170	Monday, December 1, 201
Government	Germany	Velo	None	1006	120	350	352100	0	352100	261560	90540	Sunday, June 1, 201
Government	France	VTT	None	1527	250	350	534450	0	534450	397020	137430	Sunday, September 1, 201
Government	France	Amarilla	None	2750	260	350	962500	0	962500	715000	247500	Saturday, February 1, 201
Government	Mexico	Carretera	Low	1210	3	350	423500	4235	419265	314600	104665	Saturday, March 1, 201
Government	Mexico	Carretera	Low	1397	3	350	488950	4889.5	484060.5	363220	120840.5	Wednesday, October 1, 201
Government	France	Carretera	Low	2155	3	350	754250	7542.5	746707.5	560300	186407.5	Monday, December 1, 201
Government	France	Paseo	Low	2155	10	350	754250	7542.5	746707.5	560300	186407.5	Monday, December 1, 201
Government	Canada	VTT	Low	943.5	250	350	330225	3302.25	326922.75	245310	81612.75	Tuesday, April 1, 201
Government	Mexico	VTT	Low	1397	250	350	488950	4889.5	484060.5	363220	120840.5	Wednesday, October 1, 201
Government	Canada	Carretera	Low	2852	3	350	998200	19964	978236	741520	236716	Monday, December 1, 201
Government	Canada	Paseo	Low	2852	10	350	998200	19964	978236	741520	236716	Monday, December 1, 201
Government	Germany	Velo	Low	2966	120	350	1038100	20762	1017338	771160	246178	Tuesday, October 1, 201
Government	Germany	Velo	Low	2877	120	350	1006950	20139	986811	748020	238791	Wednesday, October 1, 201
Government	Germany	VTT	Low	2877	250	350	1006950	20139	986811	748020	238791	Wednesday, October 1, 201
Government	United States of America	VTT	Low	266	250	350	93100	1862	91238	69160	22078	Sunday, December 1, 201
Government	Mexico	VTT	Low	1940	250	350	679000	13580	665420	504400	161020	Sunday, December 1, 201
Government	Germany	Amarilla	Low	2966	260	350	1038100	20762	1017338	771160	246178	Tuesday, October 1, 201

- **Defining Metrics:** Key Performance Indicators (KPIs) such as *Total Profit* and *Total Revenue* were established using DAX measures to provide management with quantifiable business insights.
- **Locate the Data:** Look at the **Data pane** on the far-right side of your screen. You should see the financials table.
- **Create "Total Revenue":**
 - Right-click the financials table name.
 - Select **New measure**.
 - In the formula bar at the top, type the following:
$$\text{Total Revenue} = \text{SUM}(\text{financials}[\text{Sales}])$$
 - Press **Enter**.

- **Create "Total Profit":**
 - Right-click the financials table name again.
 - Select **New measure**.
 - In the formula bar at the top, type the following:

Total Profit = SUM(financials[Profit])

- Press **Enter**.
- **Visualize**
 - Add card to visualize total profit and total revenue

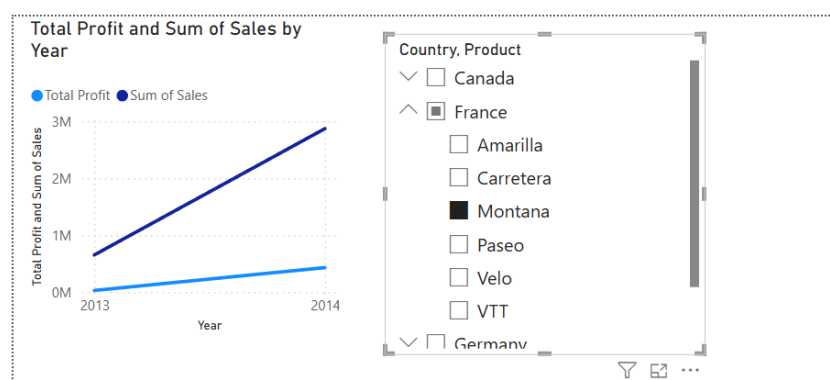


- **Designing the Dashboard:** Visuals were selected based on their ability to support business monitoring:
- **Visualizing Trends**
 - **Action:** The Line Chart icon in the Visualizations pane was clicked.
 - **Field Mapping:**
 - Dragged Date to the X-axis.
 - Dragged Sales to the Y-axis.

Adding Interactivity with Slicer

- **Action:** The Slicer icon in the Visualizations pane was clicked.
- **Field Mapping:** Dragged Country and Product into the Field.

This enables the "drill-down" feature. When a specific country in the slicer is clicked, the KPI cards and the Line Chart will automatically update to show data only for that country.



The dashboard was used to perform data drilling, demonstrating how management can filter through complex datasets to isolate underperforming products or regions, thereby facilitating data-driven strategic planning.

- **Trend Analysis:** A Line Chart was implemented to visualize revenue fluctuations over time, providing a clear view of business seasonality.
- **Interactive Filtering:** A Slicer was added to the dashboard, enabling dynamic filtering by Country and product. This demonstrated Power BI's capability to perform ad-hoc analysis, allowing users to isolate data points for deeper operational investigation.

CONCLUSION

This lab demonstrated how Power BI acts as a strategic asset for business organizations. By providing a centralized, interactive view of operational performance, it empowers management to make informed, data-driven decisions that improve efficiency and organizational outcomes.

5. Familiarization of Data Visualization by Microsoft Power BI.

THEORY

Business organizations rarely rely on a single data source. Effective Business Intelligence (BI) requires integrating diverse data streams, such as Excel spreadsheets, SQL databases, and web-based data, into a unified model. This lab explores how Power BI handles data from multiple origins, applies transformations to ensure data consistency, and visualizes complex, integrated datasets to provide a holistic view of organizational operations.

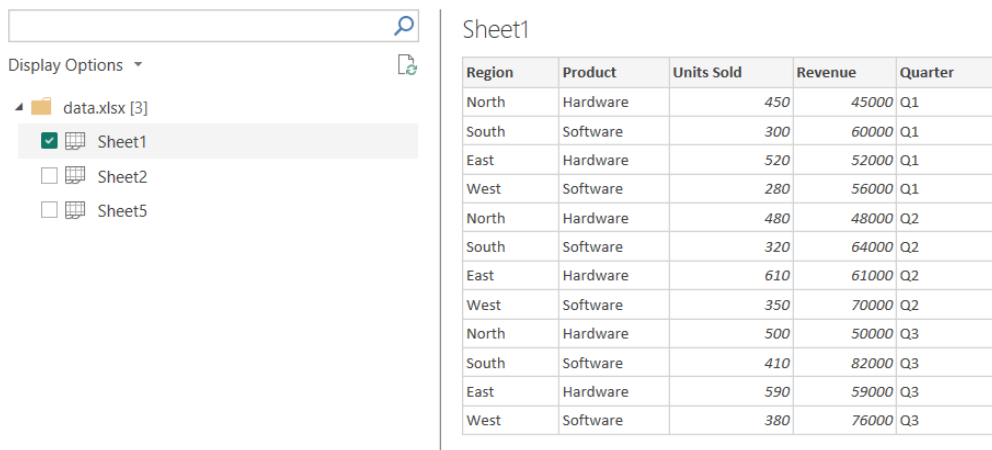
PROCEDURE

To familiarize ourselves with Power BI's data visualization capabilities, data was ingested using four distinct methods:

1. Data Import from MS-Excel:

- Selected **Get Data** from the Home ribbon and chose **Excel Workbook**.
- Navigated to the local file directory and selected the target spreadsheet.
- Verified the data structure in the Navigator pane and clicked **Load** to bring the data into the Power BI model.

Navigator



The screenshot shows the Power BI Navigator pane on the left and a data preview table on the right. The Navigator pane displays a folder named 'data.xlsx [3]' with three sheets listed: 'Sheet1' (selected with a checkmark), 'Sheet2', and 'Sheet5'. The data preview table, titled 'Sheet1', contains 12 rows of data with 5 columns: Region, Product, Units Sold, Revenue, and Quarter.

Region	Product	Units Sold	Revenue	Quarter
North	Hardware	450	45000	Q1
South	Software	300	60000	Q1
East	Hardware	520	52000	Q1
West	Software	280	56000	Q1
North	Hardware	480	48000	Q2
South	Software	320	64000	Q2
East	Hardware	610	61000	Q2
West	Software	350	70000	Q2
North	Hardware	500	50000	Q3
South	Software	410	82000	Q3
East	Hardware	590	59000	Q3
West	Software	380	76000	Q3

2. Using Sample Data:

- Selected the **"Try a sample dataset"** option directly from the Power BI canvas.
- Selected the **"Financials"** table, allowed Power BI to automatically detect data types, and clicked **Load** for immediate visualization usage.

Two ways to use sample data



Take a tutorial online

Learn how to create a robust report, step-by-step.

[Launch tutorial](#)



Experiment on your own

To start creating visuals on your own, load sample data.

Load sample data

3. Data Entry via Blank Table:

- Selected **Enter Data** from the Home ribbon.
- Manually typed or pasted specific, small-scale datasets directly into the **Create Table** grid.
- Named the table and clicked **Load** to integrate it as a custom manual data source.

Create Table

	SN	Product	Price	+
1	1	Mouse	500	
2	2	Keyboard	1000	
3				
+				

	SN	Product	Price
1	1	Mouse	500
2	2	Keyboard	1000

4. Data Import from SQL Server:

- Selected **Get Data** > **More...** > **SQL Server**.
- Input the **Server** name and **Database** name credentials.
- Selected the relevant tables from the database navigator and clicked **Load** to establish a live connection to the server.

SQL Server database

Server

localhost\SQLEXPRESS

Database (optional)

Data Connectivity mode

☒ Import

☐ DirectQuery

▸ Advanced options

OK

Cancel

Navigator

Display Options ▾

localhost\SQLEXPRESS [1]

CollegeDB [1]

☒  Students

Students

Id	Name	Age
1	Swami Sureshananda	20
4	Suresh Dahal	26

CONCLUSION

This lab successfully demonstrated Power BI's flexibility in connecting to diverse data sources. By importing data from Excel, SQL Server, manual entry, and built-in samples, we established a unified model ready for visualization. This experiment proves that Power BI is an effective tool for consolidating fragmented data into meaningful, actionable insights.